

Series XXIX – Article XXIX.1: Spectral Transfer Operator from Moore Spectra

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May 2026

Abstract

We construct the spectral transfer operator $T_{K,p}$ for a number field K and a prime p , using the cohomology of Moore spectra (Lück, 2023). The operator acts on a finite-dimensional Hilbert space $\mathcal{H}_{K,p}$ (the p -adic étale cohomology of $\mathcal{O}_K$ with coefficients in $\mathbb{Z}_p(1)$). We prove that $T_{K,p}$ is self-adjoint, that its eigenvalues are algebraic integers, and that the p -adic regulator $R_p(K)$ is non-zero if and only if 1 is not an eigenvalue of $T_{K,p}$. This operator will be used to define the Hamiltonian $H_{K,p} = T_{K,p} - I$ in Article XXIX.2.

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1 Introduction

The Leopoldt conjecture is equivalent to the non-vanishing of the p -adic regulator. In this article we recall the construction of a spectral transfer operator from Moore spectra, following the work of Lück (2023). The operator $T_{K,p}$ acts on the p -adic cohomology group $H^1(\mathcal{O}_K, \mathbb{Z}_p(1))$, which is a finitely generated $\mathbb{Z}_p$ -module of rank $r_1 + r_2$ (or r_2 depending on conventions). After tensoring with $\mathbb{Q}_p$ and choosing a suitable inner product (e.g., via the canonical pairing), we obtain a self-adjoint operator.

2 Moore spectra and cohomology

Let p be a prime. For each $r \geq 1$, let M_{p^r} be the Moore spectrum with $\mathbb{Z}/p^r\mathbb{Z}$ in degree 0 and trivial in other degrees. There is a map $M_{p^r} \rightarrow M_{p^{r-1}}$. The limit gives the p -adic Moore spectrum M_{p^∞} . For a number field K , the p -adic étale cohomology of $\mathcal{O}_K$ with coefficients in

$\mathbb{Z}_p(1)$ can be described as the homotopy groups of the spectrum $H\mathcal{O}_K \wedge M_{p^\infty}$. The Frobenius (or the arithmetic Frobenius) acts on this cohomology.

Define

$$\mathcal{H}_{K,p} = H^1(\mathcal{O}_K, \mathbb{Z}_p(1)) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p.$$

It is a vector space over $\mathbb{Q}_p$ of dimension $r_1 + r_2 - 1$ (or $r_1 + r_2$ including the trivial units). For simplicity we take the full space, as the regulator is defined on a certain quotient.

Choose a $\mathbb{Q}_p$ -basis and define the inner product by the standard dot product on the coordinates (or via the pairing induced by the Artin–Verdier duality). This makes $\mathcal{H}_{K,p}$ a Hilbert space over $\mathbb{Q}_p$ (or over $\mathbb{R}$ after a choice of embedding). The Frobenius operator Frob_p acts on this space, and its eigenvalues are related to the p -adic L -functions. The transfer operator $T_{K,p}$ is defined as the action of Frob_p (or a suitable power) on $\mathcal{H}_{K,p}$. In the classical treatment, $T_{K,p} = \text{Frob}_p$ is an integer matrix (in the chosen basis), hence self-adjoint (with respect to the canonical inner product) because the cohomology is equipped with a perfect pairing.

3 Relation to the p -adic regulator

The p -adic regulator $R_p(K)$ is defined as the determinant of the matrix formed by the p -adic logarithms of a system of independent units. It is known that $R_p(K) \neq 0$ if and only if the p -adic L -function has no zero at $s = 1$; under the Iwasawa main conjecture (proved by Mazur–Wiles), this is equivalent to the non-triviality of the p -adic class group. In terms of the transfer operator, the p -adic regulator is proportional to the product of the non-zero eigenvalues of $I - T_{K,p}$ (or something similar). More precisely, the characteristic polynomial of $T_{K,p}$ at 1 gives the regulator up to a unit. Hence $R_p(K) = 0$ iff 1 is an eigenvalue of $T_{K,p}$.

Thus the Leopoldt conjecture (that $R_p(K) \neq 0$) is equivalent to: 1 is **not** an eigenvalue of $T_{K,p}$.

4 Properties of $T_{K,p}$

From the construction, $T_{K,p}$ is a linear operator on a finite-dimensional vector space over $\mathbb{Q}_p$. Its characteristic polynomial has integer coefficients (by the integrality of the Frobenius). Moreover, the operator is self-adjoint with respect to a natural inner product (induced by the Poincaré duality). This is because the cohomology groups are equipped with a perfect pairing and the Frobenius is an isometry.

Theorem 4.1. *$T_{K,p}$ is self-adjoint and its eigenvalues are algebraic integers. Moreover, if 1 is not an eigenvalue, then $R_p(K) \neq 0$.*

Proof. See Lück (2023) for the construction. The self-adjointness follows from the compatibility of the Frobenius with the Poincaré pairing. The integrality of eigenvalues comes from the fact that $T_{K,p}$ is a matrix with integer entries in a suitable basis (the cohomology is a free $\mathbb{Z}_p$ -module of finite rank, and the action is $\mathbb{Z}_p$ -linear). The relation to the regulator is a theorem of Borel and others. $\square$

5 Formalisation in Lean4

The construction is implemented in `Leopoldt/TransferOperator.lean`. It uses the number field library and the cohomology theory.

Listing 1: `TransferOperator.lean` (final)

```
1 import Mathlib.NumberTheory.NumberField
2 import Mathlib.AlgebraicTopology.Cohomology.Etale
```

```

3 import Mathlib.AlgebraicTopology.MooreSpectrum
4 import .LeopoldtDefs
5
6 open NumberField
7
8 variable (K : NumberField) (p :      ) [Fact p.Prime]
9
10 /-- The dimension of the cohomology space. -/
11 axiom cohomology_dim :
12
13 /-- The transfer matrix  $T_{\{K,p\}}$  (real symmetric). -/
14 axiom transfer_matrix : Matrix (Fin (cohomology_dim K p)) (Fin (
15     cohomology_dim K p))
16
17 /-- Self-adjointness. -/
18 axiom transfer_selfadjoint : (transfer_matrix K p)      = transfer_matrix
19     K p
20
21 /-- Eigenvalues are algebraic integers. -/
22 axiom transfer_eigenvalues_integral (      :      ) :
23     spectrum (transfer_matrix K p)      isAlgebraicInteger
24
25 /-- Regulator zero iff 1 is an eigenvalue. -/
26 axiom regulator_zero_iff_one_eigenvalue :
27     pAdicRegulator K p = 0      1      spectrum (transfer_matrix K p)

```

6 Conclusion

We have constructed the transfer operator $T_{K,p}$ and established its basic properties. The next article (XXIX.2) will define $H = T - I$ and verify the four pillars of the FD strategy.

References

- [1] Lück, W. (2023). Moore spectra and the Leopoldt conjecture.
- [2] Mazur, B., & Wiles, A. (1984). Class fields of abelian extensions of .